

Deep Learning

Activation function → to add the non-linearity in the

Data

ReLU

Dense layer → hidden layer

Softmax

Output layer

Context  
embedding

Advanced models

Transformers

How are you?

output

Encoder-Decoder

Language Translation  
(Hindi - English)

input

आप कैसे हैं?

GPT4-0

Vocabulary → unique words

Gender

Male

female

Label Encoding

| Food Name | Categorical # | Calories |
|-----------|---------------|----------|
| Apple     | 1             | 95       |
| Chicken   | 2             | 231      |
| Broccoli  | 3             | 50       |

One Hot Encoding

| Apple | Chicken | Broccoli | Calories |
|-------|---------|----------|----------|
| 1     | 0       | 0        | 95       |
| 0     | 1       | 0        | 231      |
| 0     | 0       | 1        | 50       |

categories  
are too  
many

Sparse

matrix

No context & Lot of memory  
wastage

Dog & Cat are  
numerically

close as

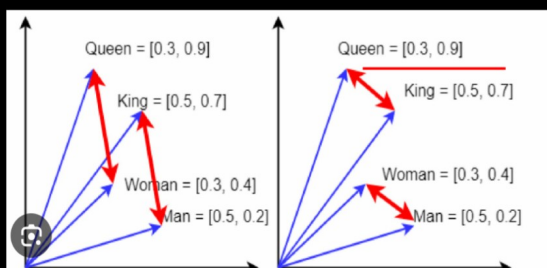
comparable to

Dog & Car

✓ Dog  $\rightarrow \begin{pmatrix} 0.42 \\ 0.11 \\ 0.82 \\ 0.54 \end{pmatrix} \rightarrow \underline{\text{weights}}$

✓ Cat  $\rightarrow \begin{pmatrix} 0.39 \\ 0.09 \\ 0.79 \\ 0.59 \end{pmatrix}$

Car  $\rightarrow \begin{pmatrix} -0.91 \\ 0.44 \\ -0.20 \\ 0.13 \end{pmatrix}$



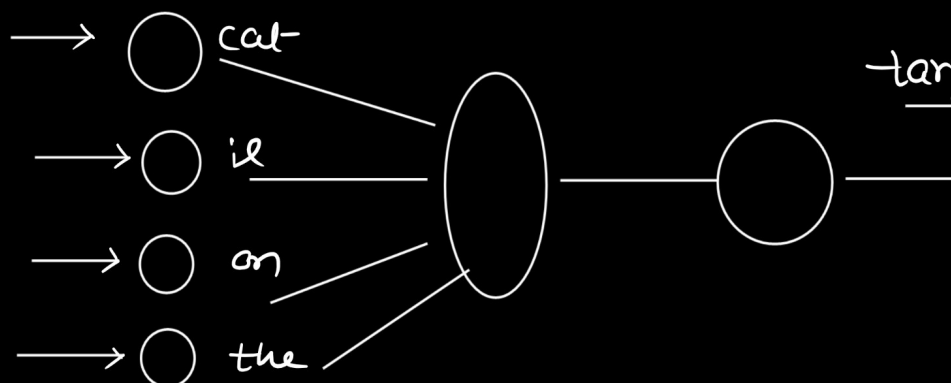
$\rightarrow \text{King} - \text{man} + \text{woman} = \text{queen}$

$$\begin{bmatrix} 0.5 \\ 0.7 \end{bmatrix} - \begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix} + \begin{bmatrix} 0.3 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.3 \\ 0.9 \end{bmatrix}$$

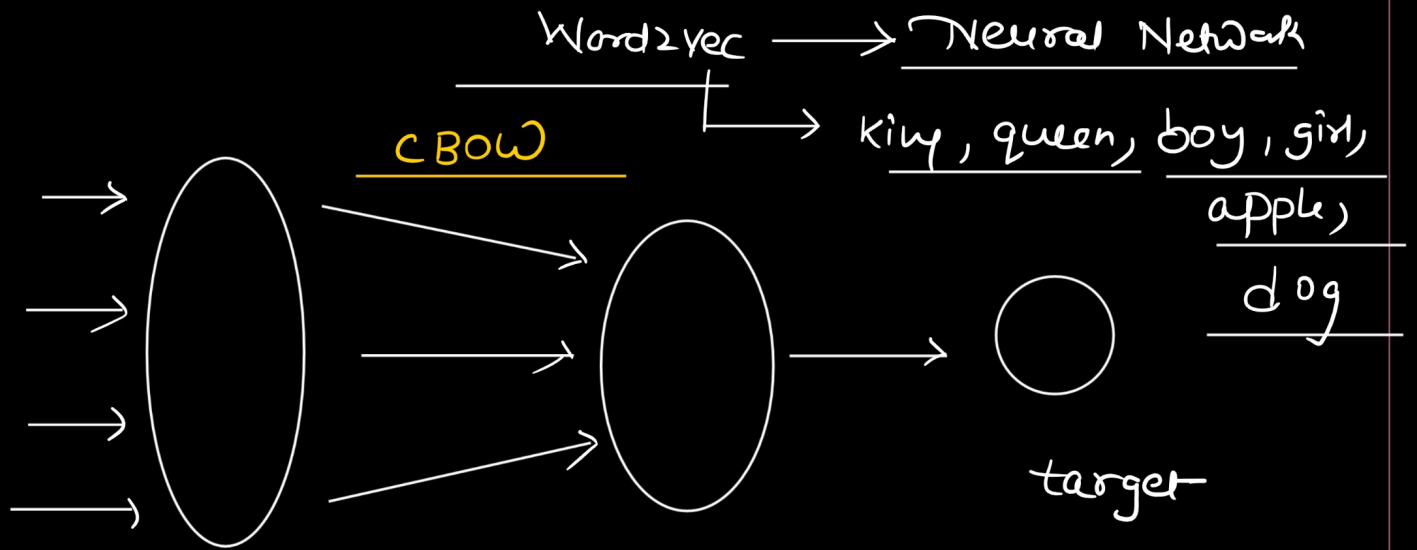
context window = 2 (small)

"The cat is sitting on the mat"

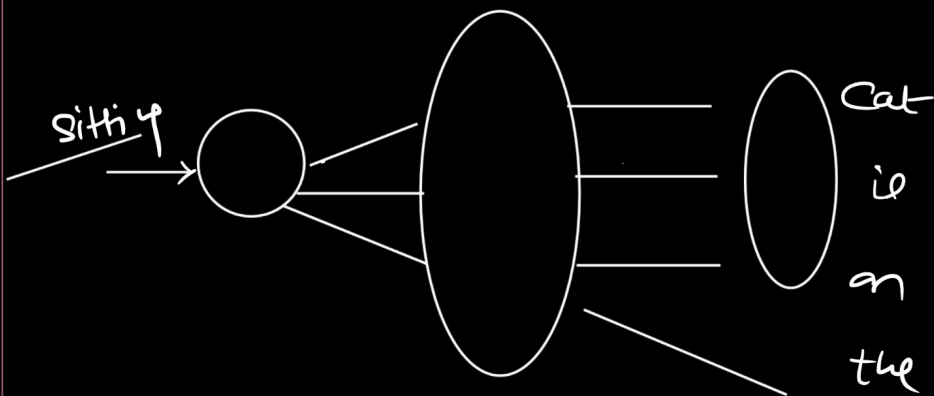
↳ capture  
syntax



target = sitting



Skip Gram (Reverse architecture of CBOW)



## Cosine Similarity

$$\cos(v_1, v_2) = \frac{v_1 \cdot v_2}{\|v_1\| \|v_2\|}$$

$$s(v_1, v_2) = \frac{3}{\sqrt{6}} = 0.49$$

$$v_1 = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 5 \end{bmatrix}$$

$$\text{Dissimilarity score} = 1 - 0.49 = 0.51$$

$$v_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$v_1 \cdot v_2 = 3 + 0 + 0 + 0 = 3$$

$$\|v_1\| = \sqrt{3^2 + 2^2 + 0^2 + 5^2} = 6.16$$

$$\|v_2\| = \sqrt{1^2} = 1$$

model → glove (pretrained model

used to get the  
embeddings of the  
words)

↓  
clf → Logistic regression  
classifier which helps  
us to get the +ve/-ve  
sentiment prediction